

Litesh Perumalla

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Summary

AI Engineer pursuing a master's in data science with expertise in building scalable machine learning systems, retrieval-augmented generation (RAG) pipelines, and multi-agent architectures. Proficient in Python, SQL, and deep learning frameworks including TensorFlow and PyTorch. Adept at engineering data pipelines, fine-tuning large language models, and integrating AI-driven solutions into enterprise applications. Passionate about leveraging advanced algorithm design, prompt engineering, and cloud technologies to architect innovative, production-ready AI solutions.

EDUCATION

University of North Texas | *Master of Science, Data Science (GPA: 3.9)*

Jan 2024 - Dec 2025

EMPLOYMENT EXPERIENCE

crypt0nest.io | *AI/ML Research & Product Intern (Team Lead)*

May 2025 - Aug 2025

- Designed and executed a baseline machine learning pipeline for a Crypto Portfolio Predictive System using Python and time-series forecasting techniques, ensuring seamless data integration from various APIs while demonstrating strong algorithmic thinking.
- Engineered volatility-based features and conducted back-testing with mean-variance optimization to strengthen model robustness.
- Integrated real-time and historical crypto data by managing NOSQL database systems, emphasizing scalable data pipeline architecture.
- Orchestrated a transition to the LLM and AI agent track by designing a multi-agent architecture incorporating retrieval-augmented generation techniques and prompt engineering for automated reasoning.
- Constructed and deployed an MCP server and client solution interfacing with LLMs, APIs, and databases, demonstrating expertise in model integration and algorithm design.
- Researched **hybrid AI systems** by blending machine learning predictions with LLM-driven agents, along with cloud technologies on Google Cloud Platform.
- Directed a multi-agent system by coordinating over five AI agents and automating workflows with integrated machine learning signals, boosting decision-making efficiency by 30%.

PROJECT EXPERIENCE

Infra Mind – AI Powered Infrastructure Platform

May 2025 – Present

- Engineered an **AI driven infrastructure advisory platform** with Intelligent dashboards and real-time analytics, executive summaries, automated infrastructure assessments, multi cloud strategy, vendor lock-in analysis, migration scenario modeling, approval workflow system, budget forecasting, rollback automation, security compliance monitoring and reporting capabilities.
- Implemented sophisticated data aggregation algorithms, real-time monitoring systems, automated alert mechanisms, comprehensive audit trails, enterprise grade security measures.
- Architected a comprehensive multi-agent system with 10+ agents and built a workflow for infrastructure provisioning, compliance monitoring.
- Tools: React.js, Typescript, Next.js, python, Fast API, MongoDB, Redis, Docker, Material-UI, RESTful APIs, Azure OpenAI.

Smart Tutor AI – Personalized Teaching Support

Jan 2025 – May 2025

- Led the design and development of an **AI-powered tutoring system** that delivers course-specific responses using validated academic materials, enhancing learning Faithfulness and relevance by semantic chunking, Hybrid Retrievers.
- Developed a efficient retrieval pipeline using metadata tagging and vector search to improve content discoverability and user interaction.
- Implemented a multi-agent architecture using LangGraph to orchestrate modular agents responsible for student ID capture, query classification, course-specific routing, and answer clarification—ensuring context-aware, personalized educational support.
- Built a Streamlit-based multi-page user interface with authentication (email, Google login), per-user chat history, feedback collection and a customizable course homepage with announcements and professor links.
- Integrated LangSmith and LangFuse for real-time system evaluation, enabling continuous fine-tuning and reliability improvements.
- Tools used: RAG, Ollama, llama Index, Chroma db, Sentence-transformers, Vector Embeddings, BERT, Stream lit.

TECHNICAL SKILLS

- Deep Learning:** Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), TensorFlow, PyTorch, Keras
- Generative AI:** RAG (Retrieval-Augmented Generation), MCP, AI Agents, Vector Database, Prompt Engineering, Embeddings
- Big Data and Cloud Technologies:** Amazon Web Services, Google Cloud Platform, Azure Cloud Services, Databricks, Terraform.
- LLM:** OpenAI, Ollama, llama Index, Lang Chain, Lang Graph, Fine Tuning, Large Language Models, AI Evaluation Methodologies
- Database Management:** PostgreSQL, MongoDB, Chroma db, FAISS
- Programming Languages:** Python, R, SQL, JavaScript, TypeScript, React.js, Next.js, Micro Services.
- Version Control & DevOps:** Git, Docker, Kubernetes, CI/CD,